



Enhanced adhesion and electrical properties of TiN/Cu metallization on Si₃N₄ ceramic substrates via magnetron sputtering



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Abstract

Titanium nitride (TiN) intermediate transition layer was introduced to enhance the interfacial adhesion and electrical properties between silicon nitride (Si₃N₄) ceramic substrate and copper (Cu) metal layer using magnetron sputtering method. Stoichiometric TiN films with sheet resistance of 14.96Ω/sq and resistivity of 1.35Ωcm were successfully synthesized under N₂/Ar mixed flow ratio of 3:20. Subsequently, the deposition of highly conductive TiN/Cu metallized bilayer, exhibiting sheet resistance of 13.13mΩ/sq and resistivity of 5.57mΩcm with thickness exceeding 2μm, was performed. It was observed that the adhesion strength improved significantly from grade 0B (direct Cu/Si₃N₄) to grade 5B, maintaining stability even after annealing

at temperatures up to 400°C. This enhancement can be attributed to dual functionality of TiN, which mediates thermal expansion mismatch while facilitating the formation of chemical bonds (Ti-N/Si-N) at the interface. This mechanism effectively suppresses stress accumulation while promoting uniform copper nucleation, resulting in increased electrical energy associated with dense copper microstructure and reduced grain boundary scattering. The proposed TiN/Cu architecture offers a robust solution for the metallization of Si₃N₄ ceramics by integrating superior adhesion, low resistivity, and thermal reliability for high-performance electronic packaging applications.



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Keywords

Si₃N₄; TiN/Cu; Magnetron sputtering; Nitrogen-to-argon flow ratio

1. Introduction

With the widespread application of Si₃N₄ ceramic substrates in electronic devices and high-temperature structural materials, surface metallization has emerged as a critical technology to enhance their performance and is regarded as a prerequisite for practical implementation [[1], [2], [3], [4]]. This process not only significantly enhances electrical conductivity and thermal conductivity but also improves performance in electronic packaging and thermal management systems [5,6]. However, the metallization of Si₃N₄ ceramics faces substantial technical challenges [7,8]. The inherent chemical inertness of Si₃N₄ surfaces and the coefficient of thermal expansion (CTE) mismatch between ceramics and metals at elevated temperatures render conventional metallization methods inadequate for achieving robust interfacial bonding and long-term stability.

The active metal brazing (AMB) process is widely employed, utilizing commercial alloys that incorporate reactive elements (e.g., Ti, Zr) in conjunction with low-melting-point metals (Ag, Cu, Sn) to establish reliable connections between Si₃N₄ ceramics and metals at relatively low temperatures [[9], [10], [11]]. This method effectively mitigates surface oxide barriers through chemical reactions that form intermetallic compounds, thereby enhancing joint strength and stability. However, challenges remain in

minimizing interfacial voids and addressing post-brazing etching issues [12,13]. A range of solutions has been proposed to tackle these challenges. For instance, Xu et al. [14] achieved high-performance bonding between Si₃N₄ and Kovar alloys using AgCuTi active filler. Their findings demonstrated that the interfacial shear strength of 154.3 MPa is influenced by a competitive reaction layer formation mechanism, as validated by density functional theory (DFT) analysis of adhesion energy differences. Furthermore, enhanced strength of Si₃N₄/Inconel heterojoints from 9 MPa to 53.5 MPa was obtained through the incorporation of W/Mo refractory interlayers combined with a Cu-Si-Al-Ti filler, which was attributed to synergistic effects arising from Ti segregation and elemental diffusion [15], [16], [17].

Magnetron sputtering, a thin-film deposition technique characterized by low-temperature processing, excellent film uniformity, and precise thickness control, has primarily been utilized in capacitor technologies [18], [19], [20]. Recent advancements have broadened its application in ceramic metallization: Xin et al. successfully fabricated Ti/Mo bilayer films on Al₂O₃ ceramics through sputtering, achieving an impressive joint strength of 63.1 MPa after sintering at 1200°C for 40 min combined with AMB [21]. Mi-Song Kim et al. demonstrated that sputtered Ag-Cu-Ti layers on Si₃N₄ substrates exhibited a peel strength of 4.26 kgf/mm, surpassing the performance of AlN substrates (2.35 kgf/mm), with no interfacial failure observed under thermal shock conditions, thereby confirming their suitability for high-reliability SiC power modules [22]. Song et al. developed a vacuum-integrated sputtering/bonding system to deposit Ti/Cu films on Si₃N₄ at 250°C under uniaxial pressure of 20 MPa [23]. The Ti seed layer facilitated the formation of TiN to enhance adhesion while nanocrystalline Cu films enabled low-temperature bonding through accelerated diffusion along the (111) plane. Zhang et al. synthesized nano-twinned Ag films exhibiting strong (111) texture on Si₃N₄ using both confocal and non-confocal sputtering modes, achieving remarkable hardness (1.82 GPa) and low resistivity (2.06 μΩcm), which can be attributed to its high-density and (111) highly-oriented twinning structure in nanoscale [24].

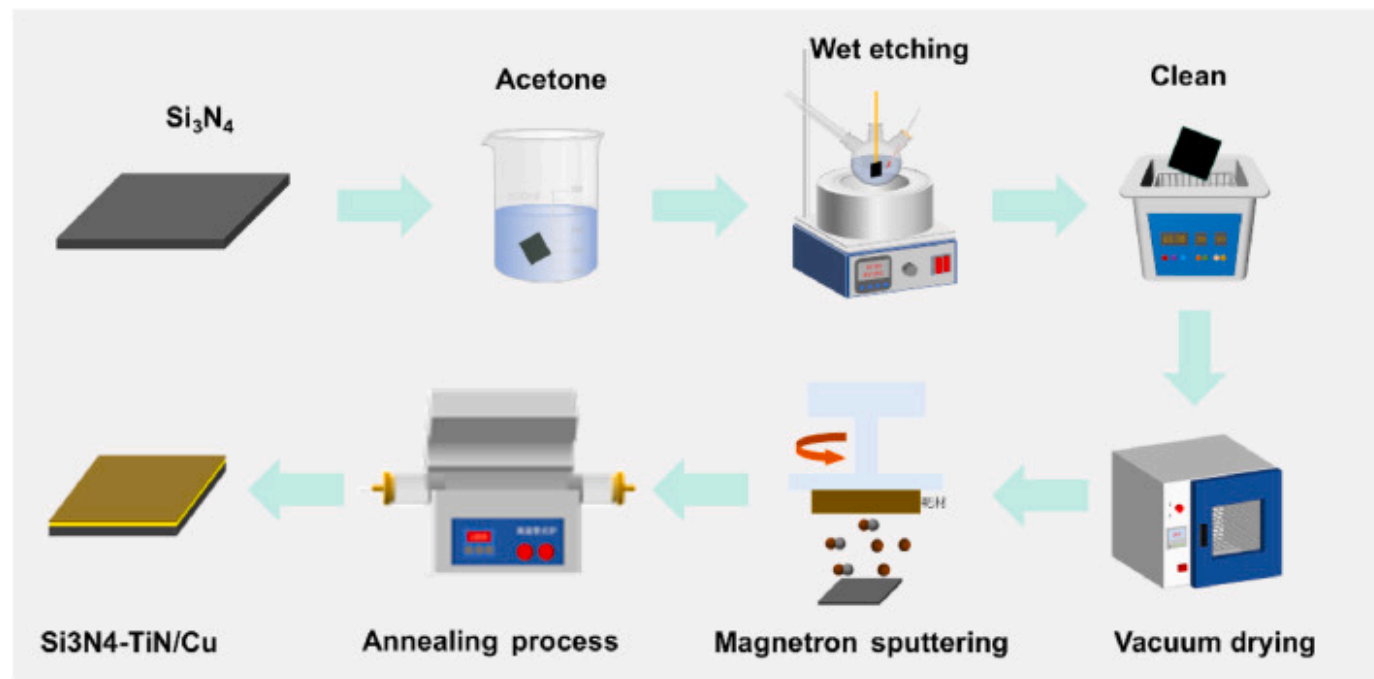
In the metallization process of Si₃N₄, Ti is commonly utilized as a transition layer due to its characteristics as an active metal and its excellent electrical conductivity. It is often integrated with techniques such as Active Metal Brazing (AMB) or Direct Bonded Copper (DBC) to facilitate subsequent metallization [25], [26], [27], [28]. However, Ti is susceptible to oxidation at elevated temperatures and can undergo interdiffusion with adjacent metal layers, which may result in diminished interfacial bonding strength.

TiN possesses a coefficient of CTE comparable to that of Ti while offering superior conductivity, high mechanical strength, corrosion resistance, and exceptional resistance to high-temperature oxidation [29,30]. The present work aims to explore the use

of TiN as an intermediate transition layer to enhance adhesion strength, optimize thin-film microstructure, and improve overall electrical performance.

2. Experiment procedure

The commercial Si₃N₄ ceramic substrate (18mm×18mm×2mm) was purchased from Guangzhou Bailing New Materials Co., Ltd. and it was the α-phase. We used acetone for cleaning to remove organic pollutants. The cleaned substrate was immersed in a phosphoric acid solution, stirred and heated for 15 min. It was then rinsed in deionized water and ultrasonically cleaned three times to remove residual phosphoric acid. After drying, the substrate was placed in a magnetron sputtering system (JGP-560a) for deposition, as shown in Fig. 1.



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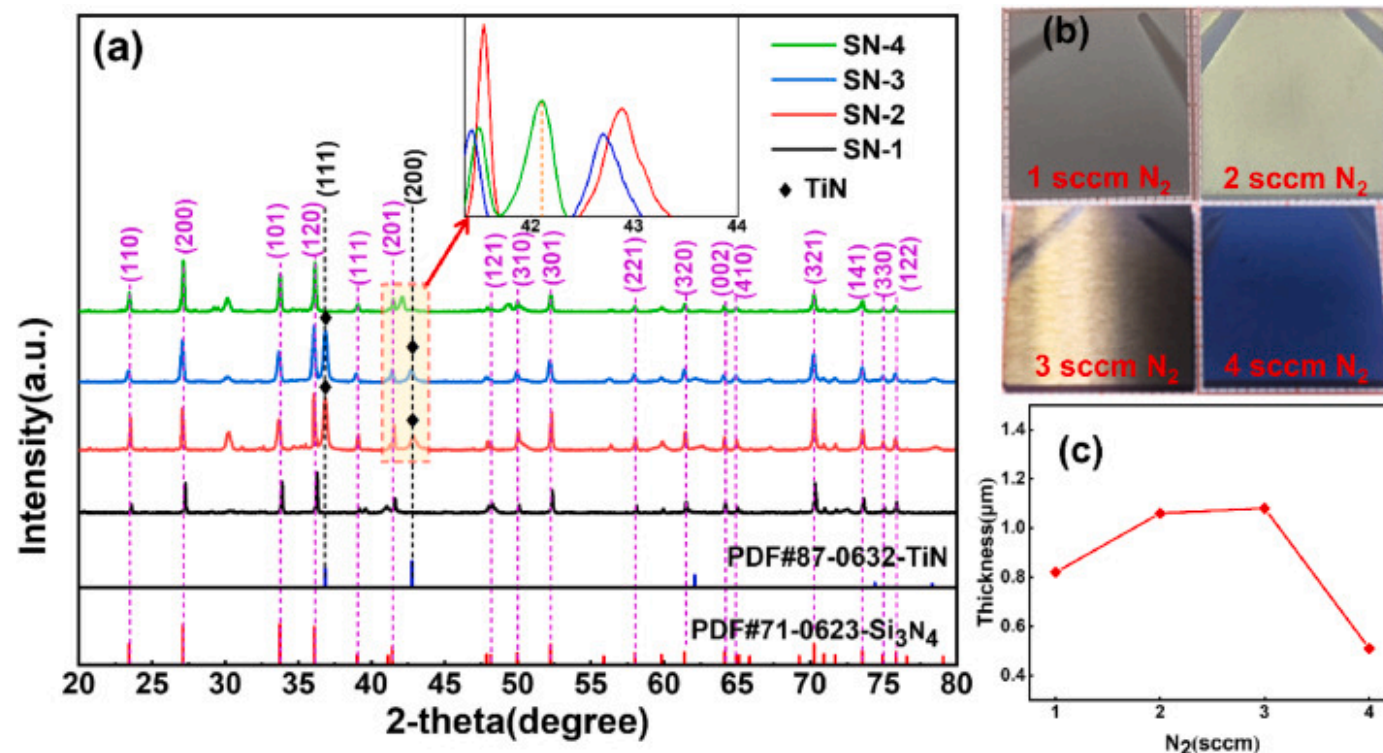
Fig. 1. Schematic of the magnetron sputtering process for TiN/Cu multilayer fabrication.

High-purity Ti(99.99%) and Cu(99.99%) targets were sequentially installed, and the pressure was evacuated to 5×10^{-4} Pa. The TiN layers were deposited with a deposition time of 60 min, a substrate temperature of 300 °C, a pressure of 1.0 Pa, an argon gas flow rate of 20 sccm, under nitrogen gas flow rates of 1, 2, 3, and 4 sccm, and as the nitrogen gas flow rate changes, the voltage and current during the deposition process remain relatively constant, and the power stabilizes at 100 W. Cu layers were deposited with a pressure of 1.5 Pa and a power of 100 W, deposition times were 10, 20, 30, and 40 min respectively. X-ray diffraction (XRD) was used for phase analysis, scanning electron microscopy (SEM) for morphological analysis, and an X-ray fluorescence coating thickness gauge (Thick 800A) for measuring the TiN and Cu thin film thickness. Electrical properties and adhesion were tested using a four-point probe resistivity tester (HPS2661) and cross-hatch tester (QFH-A), respectively.

3. Results and discussion

3.1. Phase and microstructure characterization

The XRD patterns of the TiN layers deposited under different N₂/Ar flow ratios are shown in Fig. 2(a). All films obviously exhibited characteristic crystal planes such as (101), (200), (120) from the Si₃N₄ substrate and complied with the (PDF#78–2076) standard. For sample SN-1 (1 sccm N₂), no discernible TiN diffraction peaks were observed, attributed to insufficient nitrogen supply that hindered complete reaction between Ti targets and N₂ during reactive sputtering, resulting in Ti-rich metallic phases or amorphous structures.



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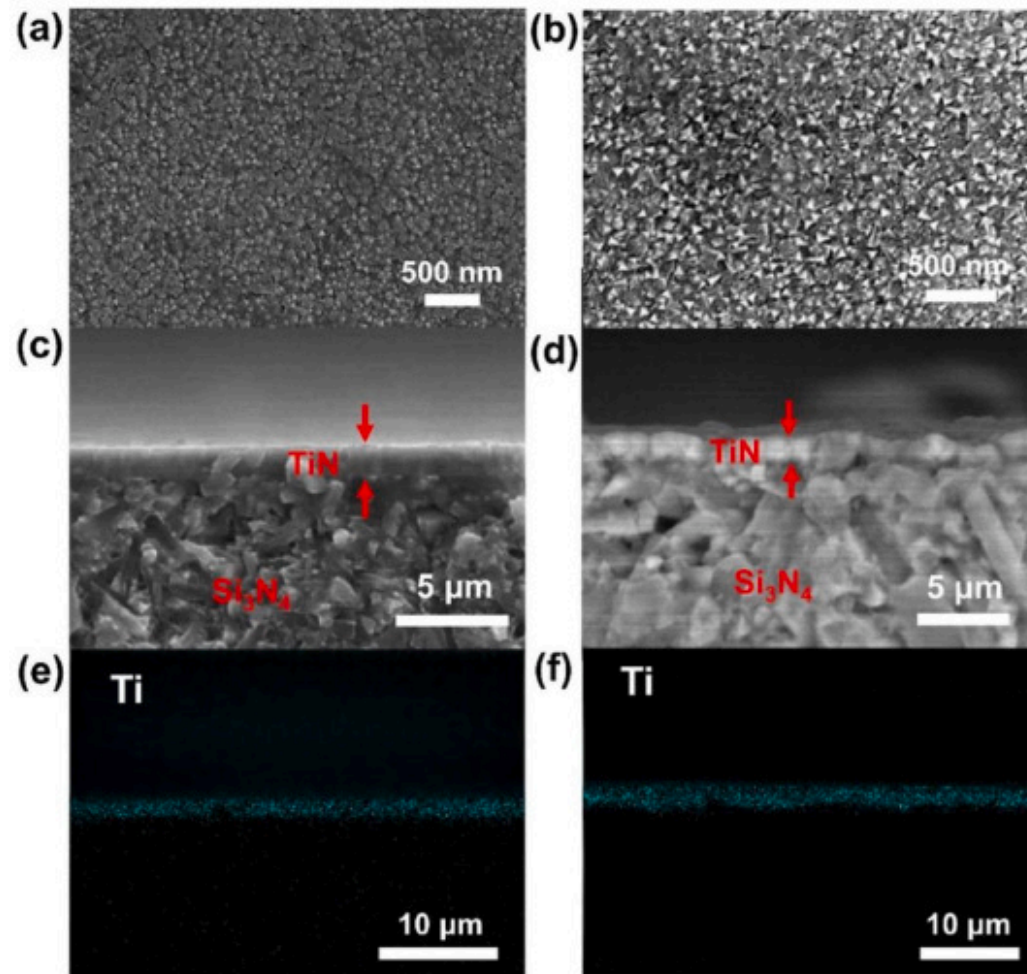
Fig. 2. (a) XRD patterns; (b) Optical images; (c) Thickness curves of TiN Films sputtered under different N₂ flow rates.

Similarly, sample SN-4 exhibited negligible TiN peaks due to excessive nitrogen flux promoting amorphous nitrogen-rich phases or hyper-stoichiometric TiN_x ($x > 1$). The diffraction peaks corresponding to the TiN(111) and (200) crystallographic planes were detected in samples SN-2 and SN-3. The (111) plane represents the minimum strain energy configuration, while the (200) plane corresponds to the minimum surface energy configuration. At low nitrogen flow rates, the (111) plane demonstrates a preferential growth orientation owing to its thermodynamic stability. However, elevated nitrogen flow modifies the internal stress distribution within the crystalline structure, inducing a progressive shift in the dominant growth orientation from the (111) plane to the (200) plane [31,32]. Simultaneously, the TiN grain size displays a non-monotonic dependence on the N₂/Ar flow ratio, exhibiting initial growth followed by subsequent reduction at higher gas ratios. This behavior stems from nitrogen

supersaturation effects at high flow rates: excessive nitrogen atoms not only inhibit grain growth through grain boundary passivation but also induce grain refinement via accelerated nucleation. The resulting lattice distortion from nitrogen incorporation elevates internal stresses and degrades crystalline quality, as evidenced by diminished long-range ordering. Additionally, the excessive nitrogen species retard surface diffusion kinetics, leading to decreased thin film deposition rates. These synergistic effects collectively drive the observed reduction in TiN grain dimensions and the deterioration of structural coherence under high nitrogen flux conditions.

Fig. 2(b) demonstrates visible substrate coloration variations across N₂/Ar conditions. At low N₂ flows (1 sccm), silver metallic Ti dominated due to limited TiN formation. Moderate nitrogen fluxes produced ϵ -Ti₂N (light yellow) and α -TiN (golden yellow), intensifying with optimized N₂ supply. Excessive N₂ (≥ 4 sccm) generated nitrogen-saturated TiN_x phases, yielding deep blue hues. Fig. 2(c) illustrates the relationship between N₂ flow rate and TiN film thickness. At a nitrogen flow rate of 1 sccm, the TiN film thickness measured 0.82 μm . The limited nitrogen supply under this condition resulted in incomplete Ti-N chemical reactions during sputtering, thereby restricting TiN growth kinetics and yielding a reduced deposition rate with thinner films. As the N₂ flow increased, enhanced nitrogen availability promoted more efficient Ti-N reactions, leading to accelerated film growth and increased thickness. The optimal N₂/Ar ratio was achieved at 3 sccm N₂, where the thickness peaked at 1.08 μm , indicating maximized reaction efficiency between Ti and N atoms. However, further increasing the N₂/Ar ratio beyond this threshold introduced nitrogen overpressure, which formed a nitride-rich layer on the Ti target surface. This passivation layer significantly reduced Ti sputtering rates, suppressed plasma reactivity, and ultimately diminished TiN growth kinetics, resulting in the formation of a thin transitional layer on the substrate.

Surface morphology observations Fig. 3(a–b) revealed uniformly distributed fine grains with high nucleation density and no macroscopic defects. Notably, as the nitrogen flow rate increases from 2 sccm to 3 sccm, the particles of SN-3 became larger and the surface became rougher. The absence of characteristic TiN trigonal facets at 2 sccm N₂ flow Fig. 3(a) suggests incomplete nitridation, and likely caused by direct deposition of unreacted Ti atoms under insufficient nitrogen supply. Cross-sectional analysis Fig. 3(c–d) demonstrated that 3 sccm N₂ flow produced TiN films with homogeneous laminar structures (consistent thickness $\approx 1.08 \mu\text{m}$) and defect-free interfacial bonding to the substrate. EDS elemental mapping confirmed uniform Ti distribution across both nitrogen conditions, with marginally higher Ti signal intensity at 3 sccm N₂ correlating with increased film thickness.



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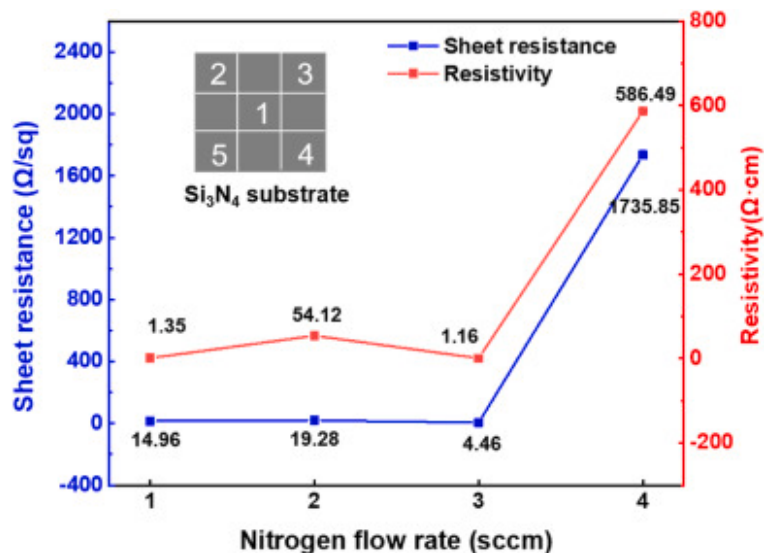
Fig. 3. SEM images:(a) Surface of SN-2; (b) Surface of SN-3; (c) Cross-section of SN-2; (d) Cross-section of SN-3; (e) Ti Element Mapping of SN-2; (f) Ti Element Mapping of SN-3.

The analysis revealed that the bonding between TiN and Si₃N₄ ceramic substrates results from the synergistic effects of multiple mechanisms. During magnetron sputtering, Ti and N atoms sputtered from the target collide with the substrate surface at high energy, forming a dense film through physical adsorption, which enhances the interfacial contact area and bonding strength.

Simultaneously, nitrogen reacts with Ti atoms during sputtering to form TiN_x compounds, with stable TiN phases generated at a nitrogen flow rate of 3 sccm, further strengthening the interfacial adhesion via chemical bonding.

3.2. Electrical properties of TiN and TiN/Cu films

As shown in Fig. 4, we measured the resistivity and square resistance at five points on the Si₃N₄ substrate and calculated the average values of the five points. The average resistivity and sheet resistance exhibit an overall trend of initially decreasing and then increasing. When the nitrogen flow rate is 1 sccm, the sheet resistance and resistivity are measured as 14.96Ω/sq and 1.35Ωcm, respectively. This behavior stems from metallic Ti-dominated conduction due to insufficient nitrogen incorporation, enabling continuous metallic bonding between Ti atoms. Increasing N₂ flow induced a transition from metallic to semiconducting characteristics, as covalent Ti-N bond formation reduced free electron density. The minimum resistivity occurred at 3 sccm N₂, where stoichiometric TiN with dense, uniform microstructure and reduced grain boundary scattering established efficient conductive pathways. However, excessive N₂ (>3 sccm) promoted amorphous or supersaturated structures with heightened electron scattering effects. As N₂ flow rate increases, collisions between Ti and N atoms become more frequent, causing a significant decrease in atomic energy. This reduction leaves insufficient energy for atoms to migrate and rearrange completely. Consequently, the crystallinity of the film decreases markedly, leading to a reduction in charge carriers within the film and a substantial increase in resistivity [33,34].



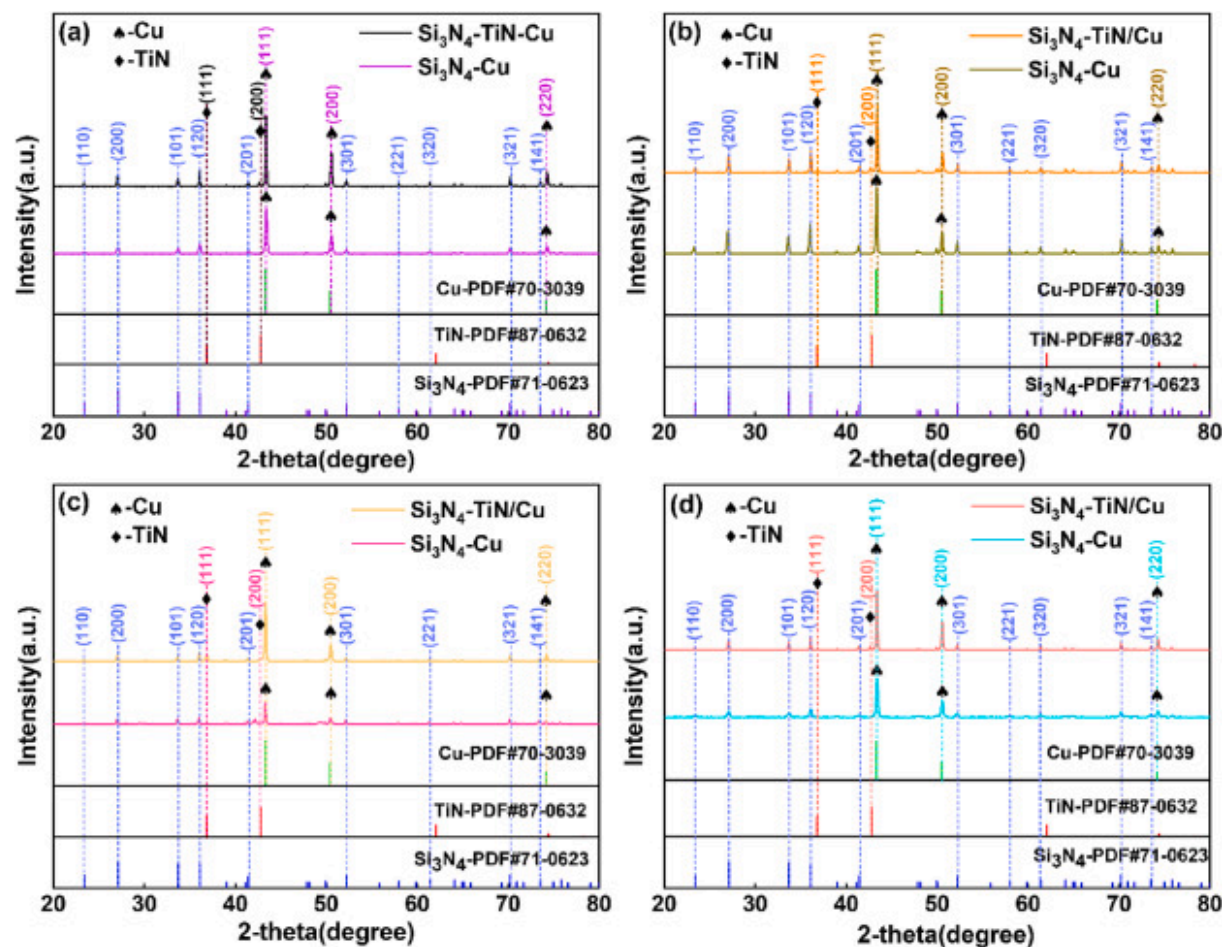
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Fig. 4. Effect of different nitrogen-argon flow ratios on sheet resistance and resistivity.

Furthermore, as shown in [Fig. 2 \(a\)](#), it can be observed that as the nitrogen gas flow rate increases to 4 sscm, the TiN (111) and (200) crystal planes disappear. An unknown peak emerges around a diffraction angle of 42°, suggesting this peak originates from nitrogen supersaturation forming excessive amounts of low-crystallinity TiN_x or non-conductive nitride phases. This process triggers the sharp increase in resistivity and the significant deterioration in electrical conductivity.

[Fig. 5](#) presents the XRD patterns of Si₃N₄-TiN/Cu and Si₃N₄-Cu samples under varying deposition durations. Both sample groups exhibited distinct (111), (200), and (220) crystallographic planes, with a predominant Cu (111) preferred orientation. Notably, the diffraction peak intensities of these planes in the Si₃N₄-TiN/Cu system consistently exceeded those of the TiN-free counterparts across all deposition times. This phenomenon can be attributed to two primary factors: deposition time-dependent crystallization dynamics and interfacial engineering by the TiN interlayer. Shorter durations (<30min) resulted in weaker diffraction signals due to incomplete crystallization, while prolonged sputtering (>60min) enhanced peak intensities but concurrently induced peak broadening and structural degradation from residual stress accumulation. Crucially, the TiN interlayer enhanced Cu deposition by optimizing surface energy modulation and lattice matching between Cu and Si₃N₄, thereby promoting epitaxial growth with superior crystallinity and intensified diffraction responses.



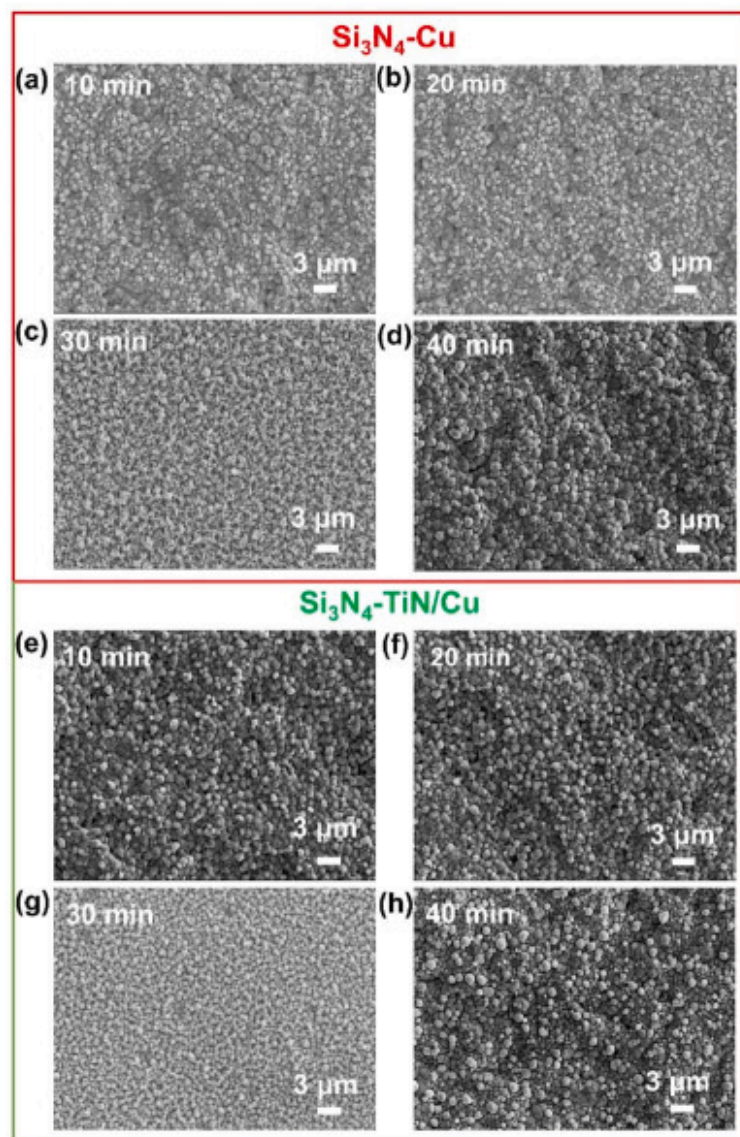
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Fig. 5. XRD patterns of Cu films at different deposition times:(a) 10min; (b) 20min; (c) 30min; (d) 40min.

Fig. 6(a–b) reveal that Si₃N₄-Cu samples deposited for 10 or 20min exhibited non-uniform size distribution and irregular spatial arrangement of surface particles. This phenomenon originates from the inherent chemical inertness and low surface energy of Si₃N₄, which induces poor wettability for Cu atoms. Such unfavorable wetting conditions promote the formation of isolated nucleation sites rather than uniform adatom spreading, leading to island-like growth modes with compromised adhesion. As show in Fig. 6(c) deposition time increased to 30min, increasingly more small spherical particles deposited onto the surface, with

partial particle coalescence occurred, resulting in irregular particle morphology, and the persistent wetting limitation formed a discontinuous film with limited densification. Further extending deposition to 40 min Fig. 6(d) triggered particle coarsening through Ostwald ripening, resulting in rough surfaces with degraded planarity.



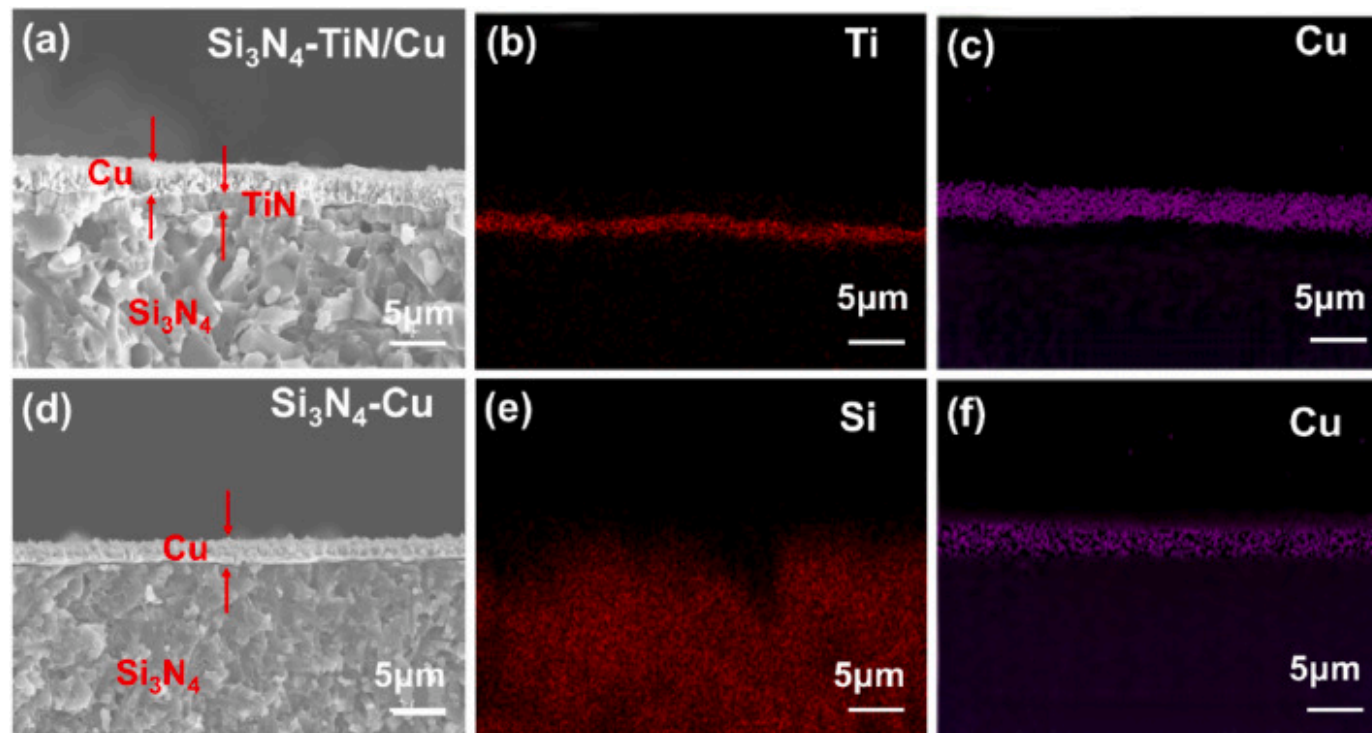
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Fig. 6. SEM images of Si₃N₄-Cu and Si₃N₄-TiN/Cu samples at different deposition times: (a–d) C; (e–h) Si₃N₄-TiN/Cu at 10–40min deposition.

In contrast, Si₃N₄-TiN/Cu samples Fig. 6(e–h) demonstrated markedly improved morphology due to the TiN interlayer's dual functionality: enhanced surface chemical activity and reduced Cu diffusion barriers. At 10–20min deposition, TiN-modified substrates achieved uniform nanoscale Cu particles with optimized spatial distribution and adhesion. After 30min, continuous, smooth Cu films with nearly complete surface coverage were observed, highlighting the role of TiN in promoting layer-by-layer growth. Even at prolonged deposition (40min), the TiN interlayer effectively suppressed excessive particle coalescence, maintaining lower surface roughness compared to TiN-free counterparts. These observations confirm that the TiN transition layer optimizes nucleation density and growth kinetics, thereby enabling high-quality Cu film deposition through interfacial energy engineering.

Fig. 7 displays cross-sectional morphology and elemental analysis of samples deposited for 30min, revealing well-defined multilayered structures with smooth fracture interfaces. The TiN interlayer exhibits high surface chemical activity, facilitating robust interfacial adhesion through dual mechanisms: mechanical interlocking with the substrate and covalent bonding via Si-N/Ti-N bond formation. Additionally, the superior wettability of TiN compared to Cu enhances compatibility with Si₃N₄, enabling uniform Cu film deposition while mitigating interfacial stress caused by CTE mismatch, thereby preventing delamination. In contrast, direct Cu deposition on Si₃N₄ resulted in moderately uniform films, but weak adhesion due to reliance solely on physical adsorption and poor wettability, rendering the films prone to cracking or interfacial failure under operational stresses. The findings underscore the critical role of TiN as a functional interlayer in achieving both enhanced adhesion and structural integrity.



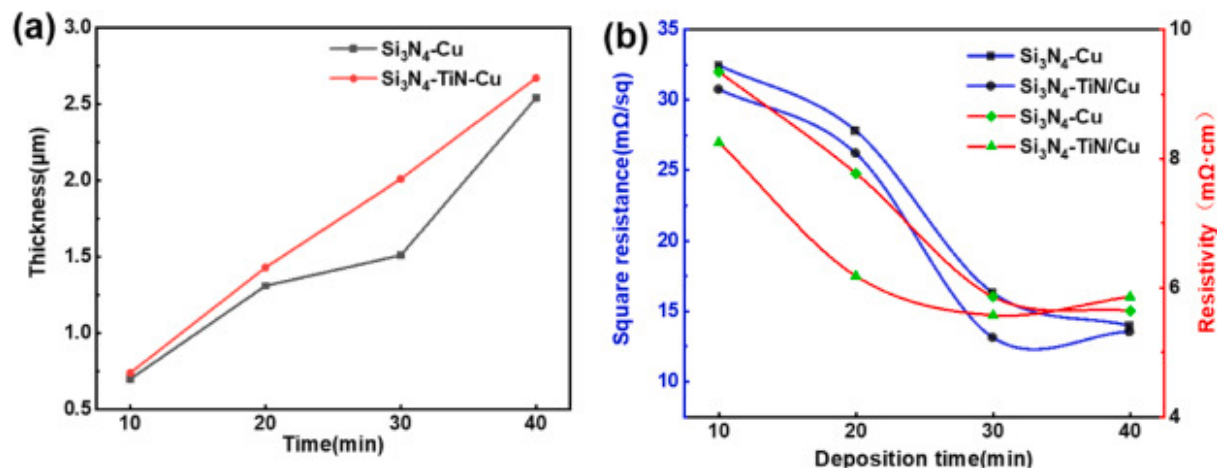
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Fig. 7. Fracture and elemental analysis of Si₃N₄-Cu and Si₃N₄-TiN/Cu samples after 30min deposition: (a) Si₃N₄-TiN/Cu; (b) Ti in Si₃N₄-TiN/Cu; (c) Cu in Si₃N₄-TiN/Cu; (d) Si₃N₄-Cu; (e) Ti in Si₃N₄-Cu; (f) Cu in Si₃N₄-Cu.

Fig. 8(a) demonstrates that Cu layer thickness in both Si₃N₄-Cu and Si₃N₄-TiN/Cu samples increases with deposition time, yet the TiN-modified system consistently achieves greater thickness under identical conditions. This disparity becomes particularly pronounced at 20–30min deposition, attributable to the enhanced surface chemical activity and wettability of TiN, which promote higher nucleation site density and accelerated Cu growth kinetics. Fig. 8(b) reveals the electrical performance evolution: sheet resistance and resistivity decrease progressively with deposition time, exhibiting rapid reduction below 30min followed by stabilization. Remarkably, Si₃N₄-TiN/Cu samples consistently outperform their TiN-free counterparts, with the most significant improvements observed between 10 and 30min. This enhancement stems from two synergistic effects: increased Cu film densification and thickness with prolonged deposition, and TiN-mediated uniform nucleation and reduced interfacial defects, as

corroborated by SEM analysis. Optimal performance was achieved with a 30min deposition for the TiN-modified system, yielding a sheet resistance of 13.13 mΩ/sq and resistivity of 5.57 mΩcm, which represent a 62% improvement over conventional Si₃N₄-Cu system and comparable performance to the Si₃N₄-TiN configuration with N₂ flow at 1 sccm for 60min [10,31].



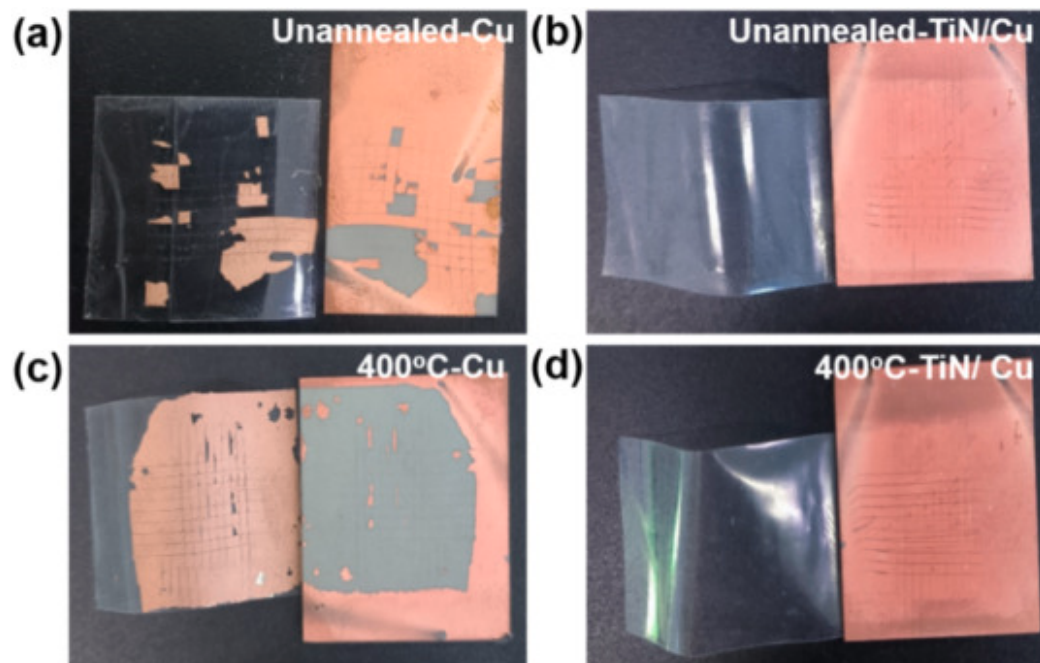
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Fig. 8. Thickness and electrical properties: (a) Cu layer thickness; (b) Sheet resistance and resistivity.

3.3. Adhesion properties TiN and TiN/Cu films

In order to enhance the adhesion of the TiN and TiN/Cu films, we carried out annealing in an argon atmosphere at 400°C for 60min. Fig. 9 presents the adhesion test results of TiN/Cu and Cu films before and after annealing. For the Cu layer, significant adhesion degradation was observed post-annealing. As shown in Fig. 9(a), the as-deposited Cu exhibited partial flaking along incision edges and localized delamination exceeding 15% but below 35% of the grid area, corresponding to a 2B rating per GB/T 9286-2021. After annealing at 400°C under reducing atmosphere Fig. 9(c), severe delamination (>90% grid area, 0B rating) occurred due to interfacial incompatibility between Cu and the inert Si₃N₄ substrate, exacerbated by Cu grain coarsening and increased porosity during annealing, which introduced structural defects and weakened interfacial cohesion.



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Fig. 9. Surface adhesion test of Si₃N₄-Cu and Si₃N₄-TiN/Cu before and after annealing: (a) unannealed-Cu; (b) unannealed-TiN/Cu; (c) 400°C-Cu; (d) 400°C-TiN/Cu.

In contrast, the TiN/Cu system Fig. 9(b–d) maintained 5B-rated adhesion (no flaking) under both conditions. The TiN interlayer functioned as a bridging phase, enhancing adhesion through three mechanisms: (1) Mitigation of thermal expansion mismatch by intermediate CTE values (TiN: $\sim 9.3 \times 10^{-6}/\text{K}$; Si₃N₄: $\sim 3.2 \times 10^{-6}/\text{K}$; Cu: $\sim 16.5 \times 10^{-6}/\text{K}$); (2) Formation of stable Ti-N and Si-N covalent bonds at the interface. As reported by Wei Feng et al., titanium nitride and silicon nitride could form a strongly bonded interface [35]. Simultaneously, a layer of ternary TiCuN nitride formed at the titanium-copper interface. This compound was relatively stable and was embedded within the copper layer. These factors collectively enhanced the interface bonding strength and improved coating adhesion; (3) Suppression of Cu grain boundary migration during annealing. It was indicated that during annealing, copper films deposited via magnetron sputtering exhibited grain growth driven by the high free energy at Cu grain boundaries [36]. In contrast, the presence of tungsten atoms in Cu-W films induced a pinning effect on Cu grain migration,

thereby conferring higher thermal stability to Cu-W films. Therefore, for the TiCuN layer formed in this study, the presence of Ti atoms may similarly exert a pinning suppression effect on Cu grain boundary migration, thereby enhancing the coating's adhesion. Therefore, during the annealing process, the synergistic effects of multiple mechanisms effectively preserved the interfacial integrity, highlighting the crucial role of TiN in facilitating robust metallization on Si₃N₄ ceramics under thermal stress.

4. Conclusion

This study demonstrates the successful deposition of TiN/Cu films as metallization seed layers on Si₃N₄ ceramic substrates, with systematic investigation of sputtering parameter effects yielding the following conclusions.

- (1) The N₂/Ar flow ratio critically governs TiN film quality, where optimized conditions (N₂:Ar=3:20, 30min sputtering, 1.0Pa pressure) produced TiN layers with sheet resistance and resistivity of 14.96Ω/sq and 1.35Ωcm, respectively. Further refinement of TiN/Cu films at 30min sputtering duration achieved superior performance: 13.13mΩ/sq sheet resistance, 5.57mΩcm resistivity, and thickness exceeding 2μm.
- (2) Adhesion analysis of Si₃N₄-Cu and Si₃N₄-TiN/Cu systems before and after annealing revealed significant interfacial differences. Post-annealing evaluation showed the adhesion grade of Si₃N₄-Cu degraded from 2B to 0B, indicating that direct Cu deposition on Si₃N₄ ceramic substrates fails to achieve robust bonding due to interfacial mismatch and insufficient chemical interaction. In contrast, the Si₃N₄-TiN/Cu system maintained stable adhesion performance without observable peeling or delamination phenomena throughout the annealing process. This superior behavior originates from the TiN interlayer's multifunctional contributions: its tailored thermal expansion coefficient bridges the mismatch between Si₃N₄ and Cu, while its chemical compatibility facilitates interfacial bonding through the formation of transitional phases. These synergistic effects enable the TiN layer to serve as an optimized seed layer for subsequent substrate metallization, effectively suppressing thermal stress accumulation and preserving interfacial integrity under thermal cycling conditions.

CRedit authorship contribution statement

Jiacheng Chen: Writing – original draft, Data curation. **Letian Bai:** Writing – original draft, Data curation. **Gang Yu:** Writing – review & editing, Methodology. **Shuai Yang:** Visualization, Formal analysis. **Yanfang Li:** Writing – review & editing, Methodology. **Guanglei Zhang:** Project administration, Methodology, Conceptualization.

Declaration of competing interest

All authors declare that they have no conflicts of interest. There are no competing financial interests or personal affiliations relevant to this research that could have influenced the conduct, interpretation, or reporting of the findings in this manuscript.

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